



# LAR Heater Pigging & Decoking Services

**Marathon Petroleum, Los Angeles Refinery**

Carson Coker #1 & #2, Wilmington DCU H-101

**Presented by Marshall Douglas**  
Director of Pigging Operations  
DSP#26098 | September 2026



# Agenda

Structured to the evaluation team's ten discussion points



- |           |                                   |           |                                           |
|-----------|-----------------------------------|-----------|-------------------------------------------|
| <b>01</b> | Pigging & decoking methodology    | <b>06</b> | Emergency response & local resources      |
| <b>02</b> | Execution plan & schedule         | <b>07</b> | Safety & risk mitigation                  |
| <b>03</b> | Operational readiness & resources | <b>08</b> | Best practices, capabilities & innovation |
| <b>04</b> | Equipment & technology            | <b>09</b> | MPC references & case studies             |
| <b>05</b> | Cleanliness verification & QA/QC  | <b>10</b> | Why USA DeBusk                            |

# Our understanding of the scope

Recurring mechanical pigging & decoking, lump sum per occurrence, mobilization and demobilization waived

| Unit            | Heater    | Passes | Per year | 5 years |
|-----------------|-----------|--------|----------|---------|
| Carson Coker #1 | Heater #1 | 2      | 4        | 20      |
| Carson Coker #1 | Heater #2 | 2      | 4        | 20      |
| Carson Coker #2 | Heater #3 | 2      | 3        | 15      |
| Wilmington DCU  | H-101     | 4      | 1        | 5       |

Source: DSP#26098 Rev 1, scope table

**4**

heaters, two sites

**12**

decokes per year

**60**

decokes over 5 years

**2,790**

project hours, 5 years



Marathon Los Angeles Refinery  
Carson delayed coker

Photo: Marathon Petroleum

# Bi-directional mechanical decoking

Every pass follows the same controlled sequence

1

## Rig-in & pressure test

Connect 300# launchers and receivers; fill and test before any launch

2

## Baseline flow test

Record pressure at a set rate for a before-and-after comparison

3

## Gauge foam run

Proves the pass is open and sets the initial pig size

4

## Scraper pig progression

Carbide-studded pigs run in both directions, stepped up  $\frac{1}{8}$ " per clean pass

5

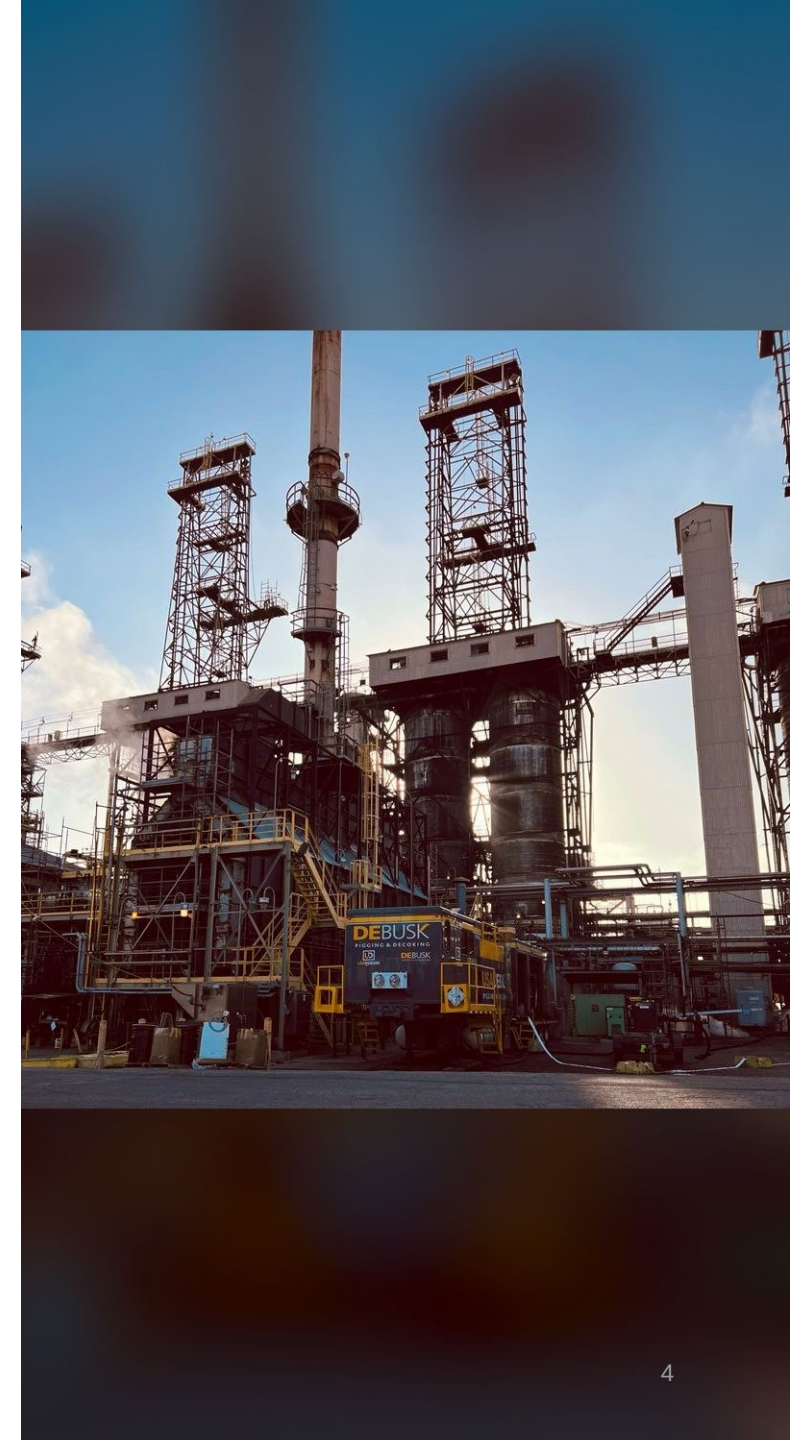
## Oversize polish

Up to tube ID +  $\frac{1}{4}$ " maximum, with real-time pressure monitoring

6

## Gauge foam & sign-off

Line-size foam witnessed by Marathon, final flow test, pass signed clean



# Pig progression plan by heater

Answers RFI item SP-2; the gauge foam run sets the starting size, the tube ID sets the ceiling

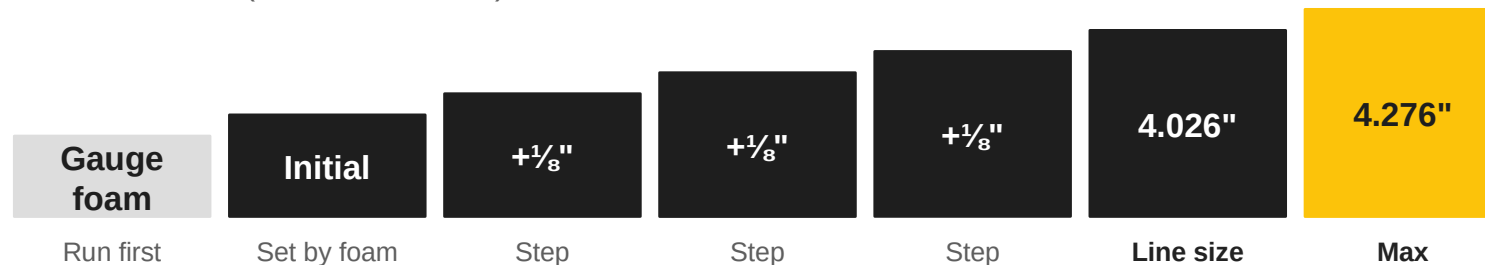
## Carson Cokers, Heaters #1, #2, #3

Governing ID 3.826" (4" Sch 80)



## Wilmington DCU H-101

Tube ID 4.026" (4.5" OD × 0.237")



### The sizing rule

**Max pig OD =  
smallest tube ID + 0.250"**

- Step up 1/8" after each clean pass
- Pig wear drives size & appendages
- Body length set for bends & mule ears
- Gauge foam sets the starting size
- Line-size foam witnessed at sign-off

# Engineered decoking pigs, matched to the coke

The most diverse selection of engineered decoking pigs in the world



*Foam, carbide-studded and worn pigs from USA DeBusk jobs*



## Urethane body, foam core

Flexes through return bends while flow bypasses the pig and carries debris away



## Durometer chosen per job

Softer compounds compress through tight bends; harder compounds remove coke aggressively



## Bald or studded

Carbide studs as standard, with alternate stud materials matched to the tube metallurgy



## Heavy-coke designs

High-durometer, fully studded pigs brought in when pig wear shows hard, heavy coke

# Heater data & coil-specific controls

| Heater        | Section    | Passes | Footage | Tube ID         | Metallurgy |
|---------------|------------|--------|---------|-----------------|------------|
| Carson Cokers | Convection | 2      | 2,268'  | 3.826"          | SS 321     |
| Carson Cokers | Radiant    | 2      | 4,032'  | 3.826" / 4.026" | A335 P9    |
| H-101         | Convection | 4      | 2,508'  | 4.026"          | A312 347H  |
| H-101         | Radiant    | 4      | 7,524'  | 4.026"          | A312 347H  |

Source: DSP#26098 Rev 1, Technical Data (coil data provided by Marathon)

## 6,300'

per Carson heater

## 10,032'

on H-101

## Grade

launcher access (H-101  
radiant: grade & 1st deck)



### Stainless coils

Water quality verified before fill;  
passivation coordinated with Marathon; no  
galvanized contact with tubes



### Connections

600# RFWN coil flanges adapted to USA  
DeBusk 300# launchers and receivers

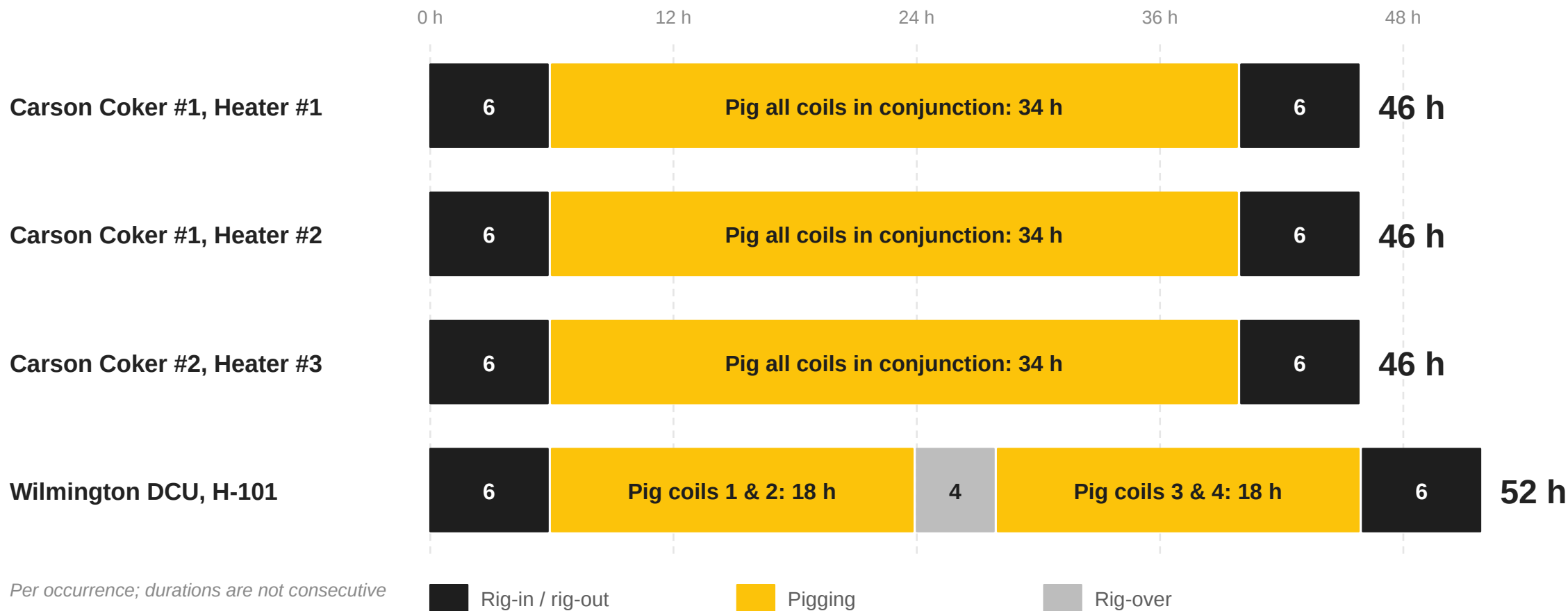


### Return bends

Pig length and flexibility selected for 180°  
returns and mule ears

# Execution plan per occurrence

Continuous 24/7 pigging on 12-hour shifts; passes pigged in conjunction



# Five-year program at a glance

Illustrative annual cadence: dates are set to Marathon's run-length and turnaround plan



**12** decokes a year

**60** decokes in 5 years

**2,790** project hours

**\$0** mob / demob

*Same crew leadership, same equipment and same procedures every visit: each decoke builds on the last one's records.*

# Crew & equipment, every occurrence



## Equipment

- 1 TriMax triple pumper
- 1 Support unit: fittings, tooling, spares
- 2 Crew trucks



## Manpower

- 2 Supervisors: 1 day, 1 night
- 3 Operators: 2 day, 1 night
- 1 Project manager

**~5%**

crew turnover, last 12 months

**24/7**

two 12-hour shifts

**100%**

training verified before mobilization

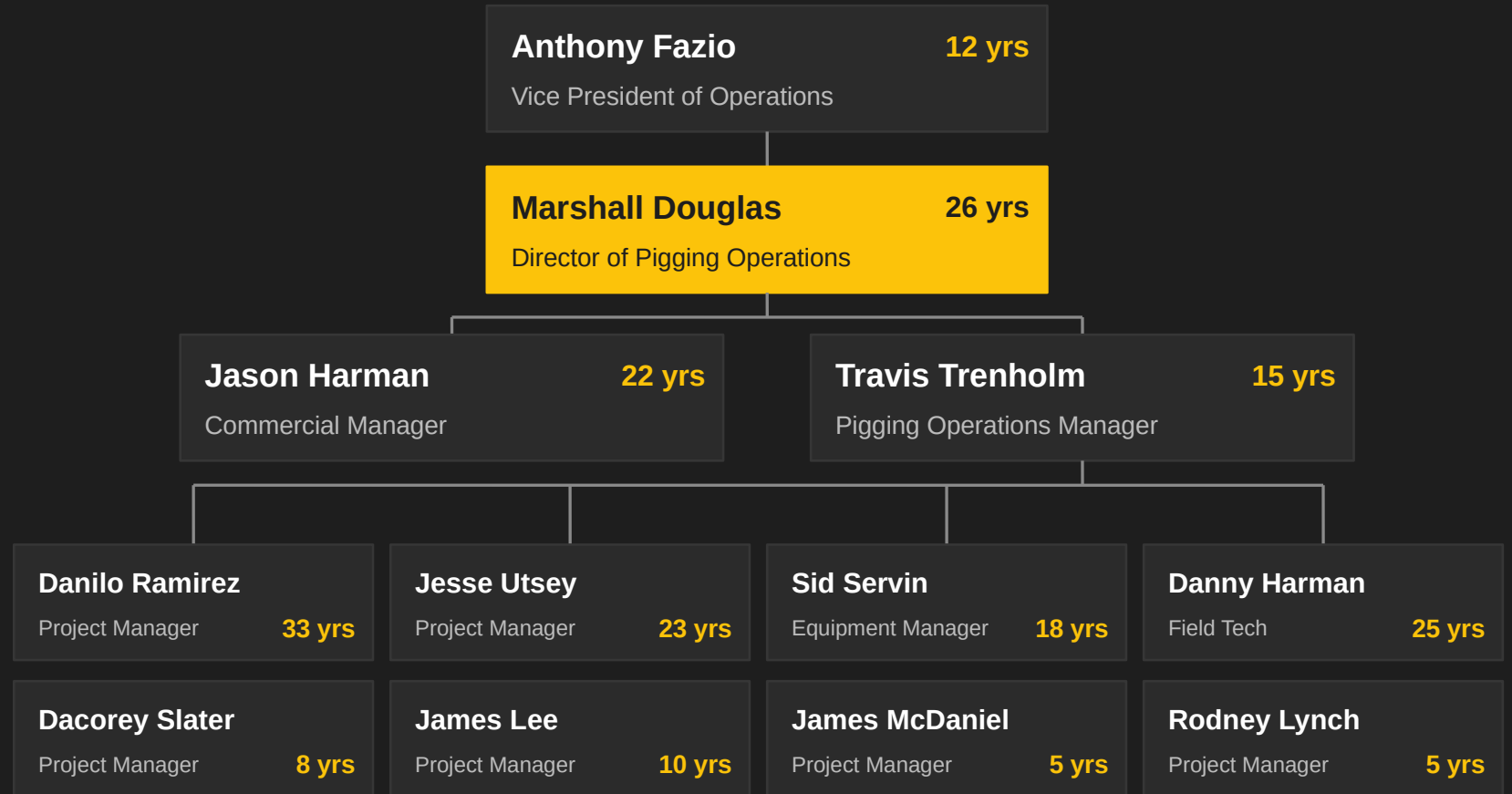
# Your LAR project team

**202**

years of combined pigging & decoking experience

**15+**

years self-performing heater pigging



# The TriMax triple pumper

6 identical TriMax units, 18 engines: one standardized fleet



## Three independent pump systems

Closed-loop, bi-directional; 50% more productive than a twin pumper



## Automated HMI control

Three 17" touchscreens, automated digital pig logs



## 150–300 psi working pressure

Pump rated 600 psi; 500–550 psi effective after valve losses



## Real-time sensor data

Coke shows as pressure spikes at the operator interface



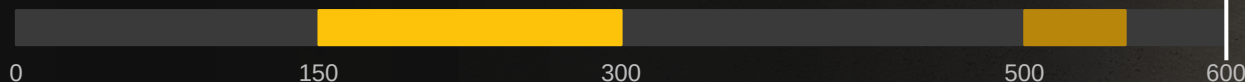
## Cummins QSL9, 333 bhp, Tier 4 Final

CARB-registered engines, California-ready

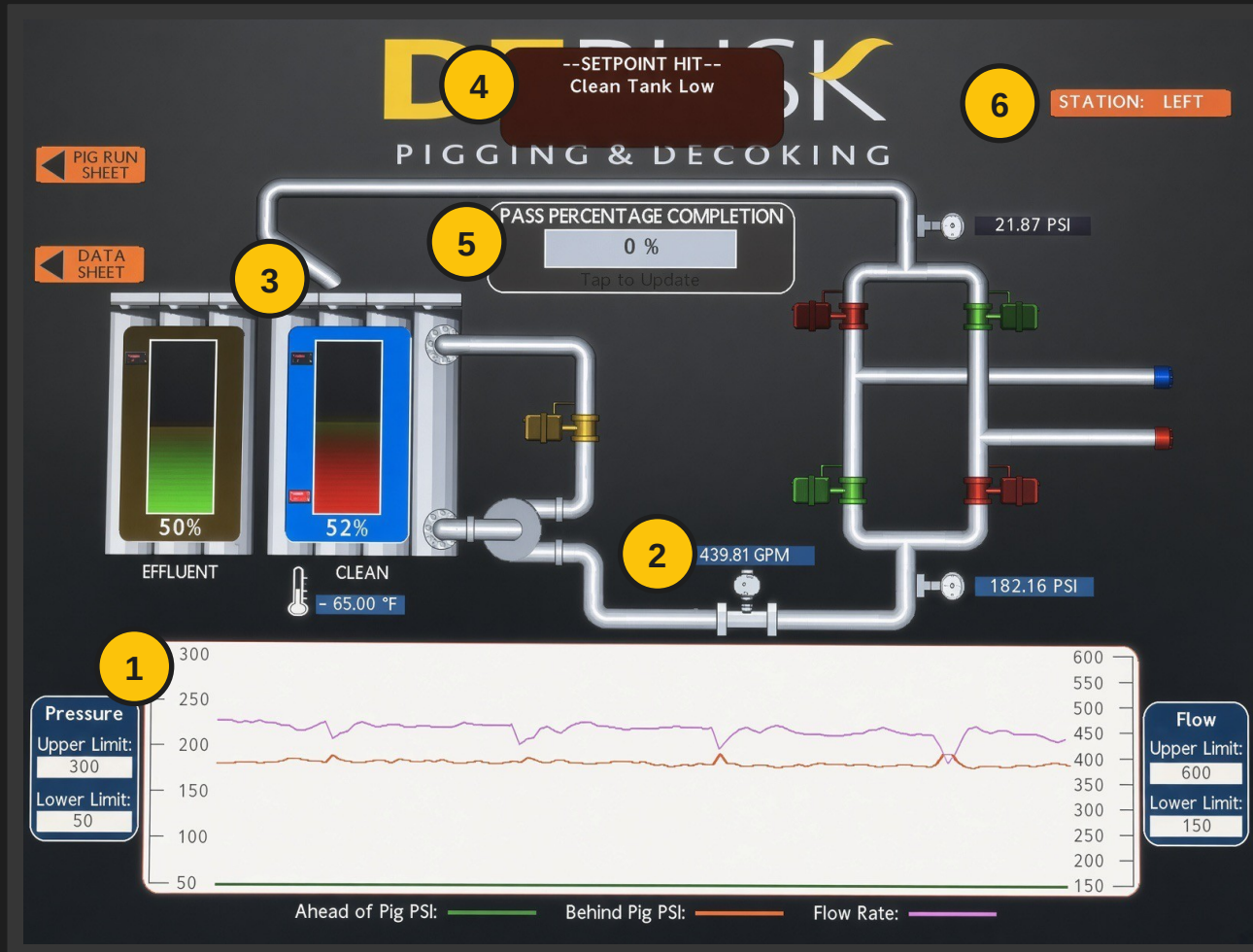
OPERATING PRESSURE, PSI

Normal pigging

Effective max



# Real-time control at every operator station



- 1 Live trends**  
Pressure ahead of and behind the pig, plus flow; coke shows as a spike
- 2 Flow & pressure**  
GPM and psi at the pump, updated continuously
- 3 Tank levels**  
Clean and effluent tanks on guided-radar indicators
- 4 Setpoint alarms**  
Upper and lower limits on pressure and flow trigger alerts
- 5 Digital pig logs**  
Pass completion, pig run sheet and data sheet recorded automatically
- 6 One station per pass**  
Left, center and right stations run independently

# California-ready, CARB-registered fleet

CARB Statewide Portable Equipment Registration (PERP); certificates shown

6

identical TriMax units, SCAQMD-ready

18

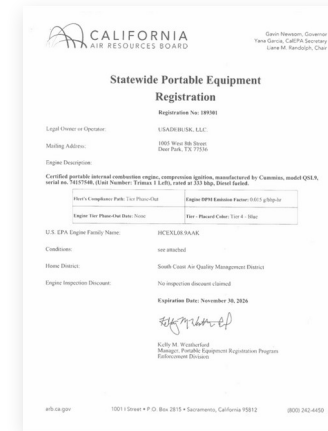
registered pumping engines

Tier 4

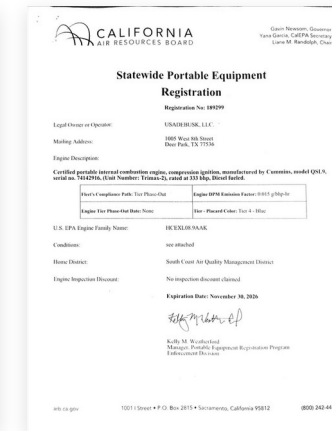
engines (blue placard)

0.015

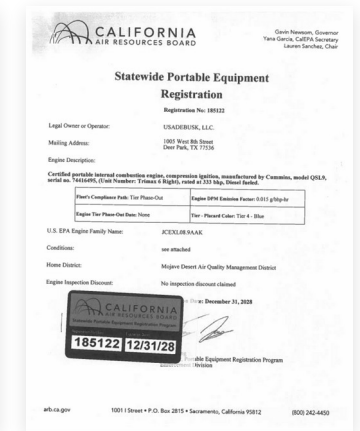
g/bhp-hr diesel PM emission factor



TriMax 1



TriMax 2



TriMax 6

CARB PERP registration certificates



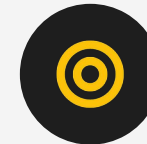
Support unit



300# launchers & receivers



Tested & certified  
hard and flex pipe



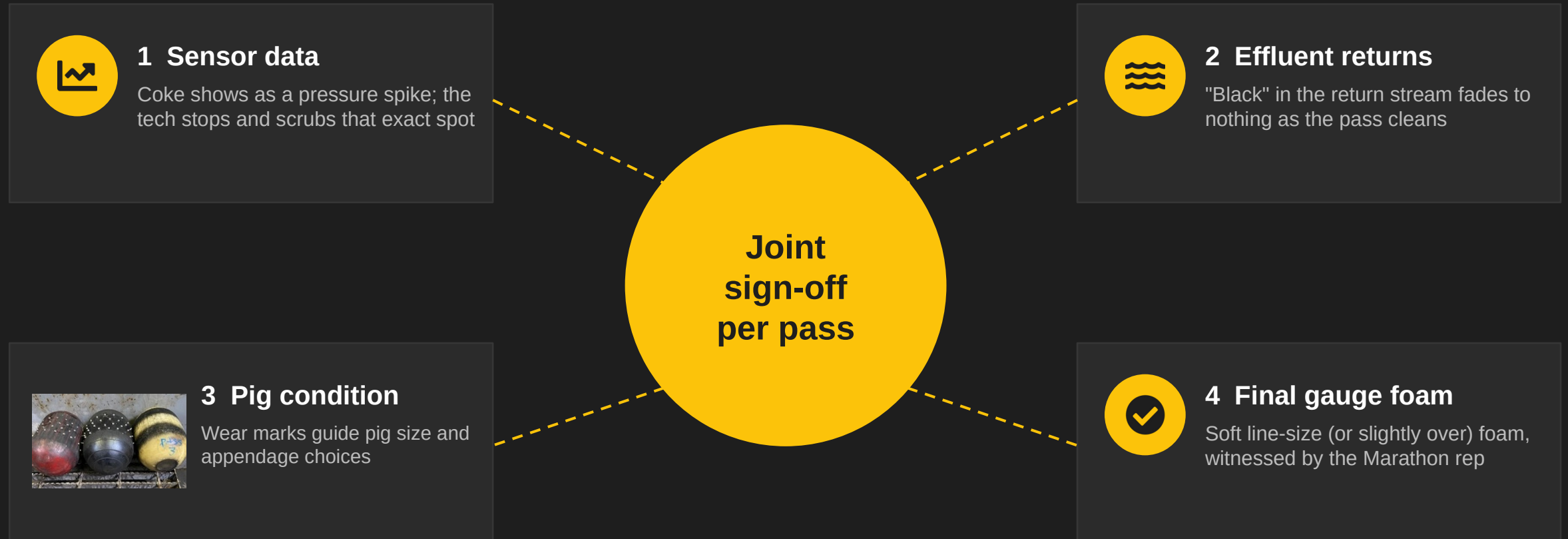
All pigs & foam  
swabs



Optional: closed-loop  
filtration

# Verifying every pass is clean

Four independent indicators, reviewed side by side with Marathon



*Proven in the field: successful sign-offs from all of our clients, and highly effective for first-attempt smart pig data.*

# QA/QC from walkdown to job book

Governed by the USA DeBusk QA/QC manual, submitted with the RFI response



## Before

- Pre-job walkdown with operations
- Coil data checked against Marathon records
- JSA, permits and isolation confirmed
- Pig progression staged by heater



## During

- Daily report to one point of contact
- Automated digital pig log per pass
- Flow tests before and after at the same rate
- JSA reviewed at every shift change



## Close-out

- Per-pass cleanliness sign-off
- Final pig size recorded per pass
- Job book delivered to Marathon
- Post-job review and lessons learned

# Ready when LAR needs us

Shops, equipment and crews in Southern California / LA Basin, with regional support

**24/7**

Dispatch and account manager, any hour

**Same day**

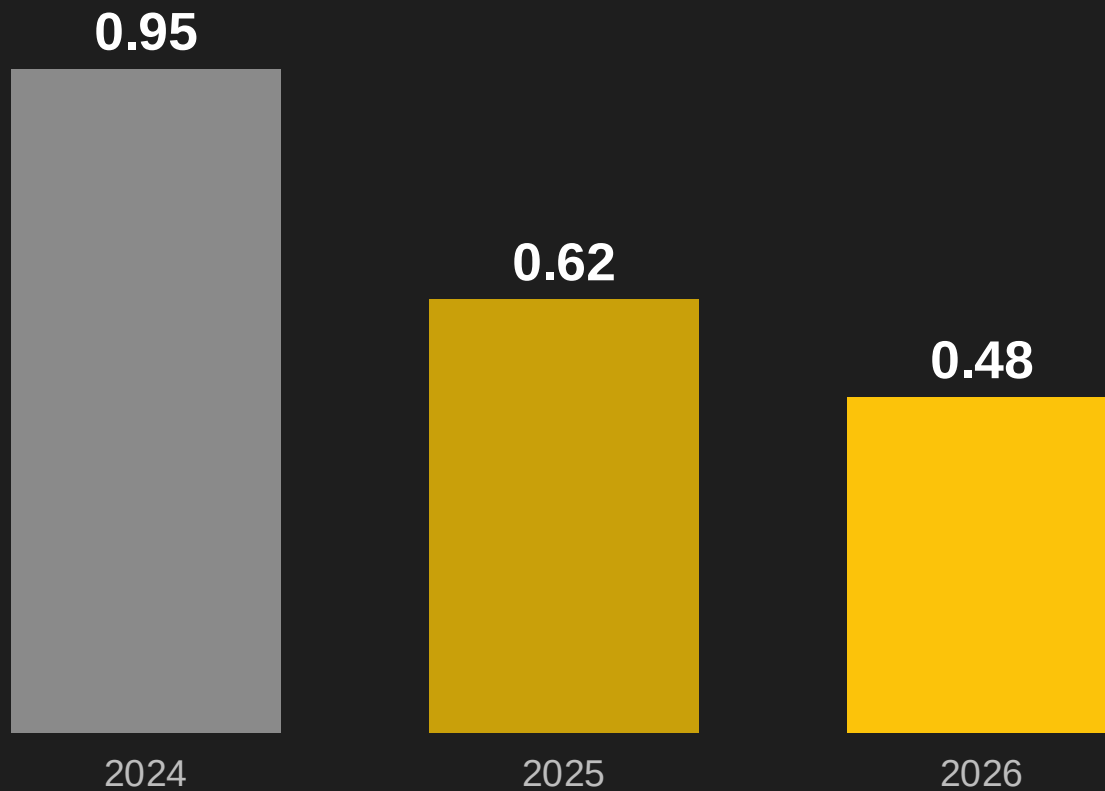
Response for local needs

**< 24 hrs**

Manpower & equipment anywhere else, from regional shops

# Safety performance

Experience Modification Rate: 1.0 is the industry baseline; lower is better



**A**

ISNetworld safety grade

**0.48**

EMR, 2026

**-49%**

EMR since 2024

**0**

fatalities in 3 years

# Risk mitigation in the field



## JSA & permits

Job-specific JSA reviewed at every shift change; daily toolbox talks



## Isolation first

Marathon's isolation and LOTO verified before any connection is made



## Pressure control

Normal 150–300 psi; over-pressure checklist above 500 psi



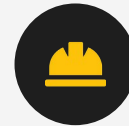
## Line of fire

Bleed to zero at every pig change; nobody in line with lids or hoses



## Hose integrity

Tested & certified hard and flex pipe; whip checks on every connection



## PPE & coil care

FRC and face shields; no galvanized metal on stainless

**Engineered for safety** Retractable cam-actuated stairs, composite work platform, LED lighting



# What sets USA DeBusk apart



## Self-performed scope

Pigging, decoking and hydroblasting with our own equipment and SCAQMD rigs: one contractor, no subs



## Largest standardized fleet

6 identical TriMax units, 18 engines, with a spare pump on board even on a 2-pass heater



## Widest pig selection

Engineered decoking pigs matched to each coil and each coke condition

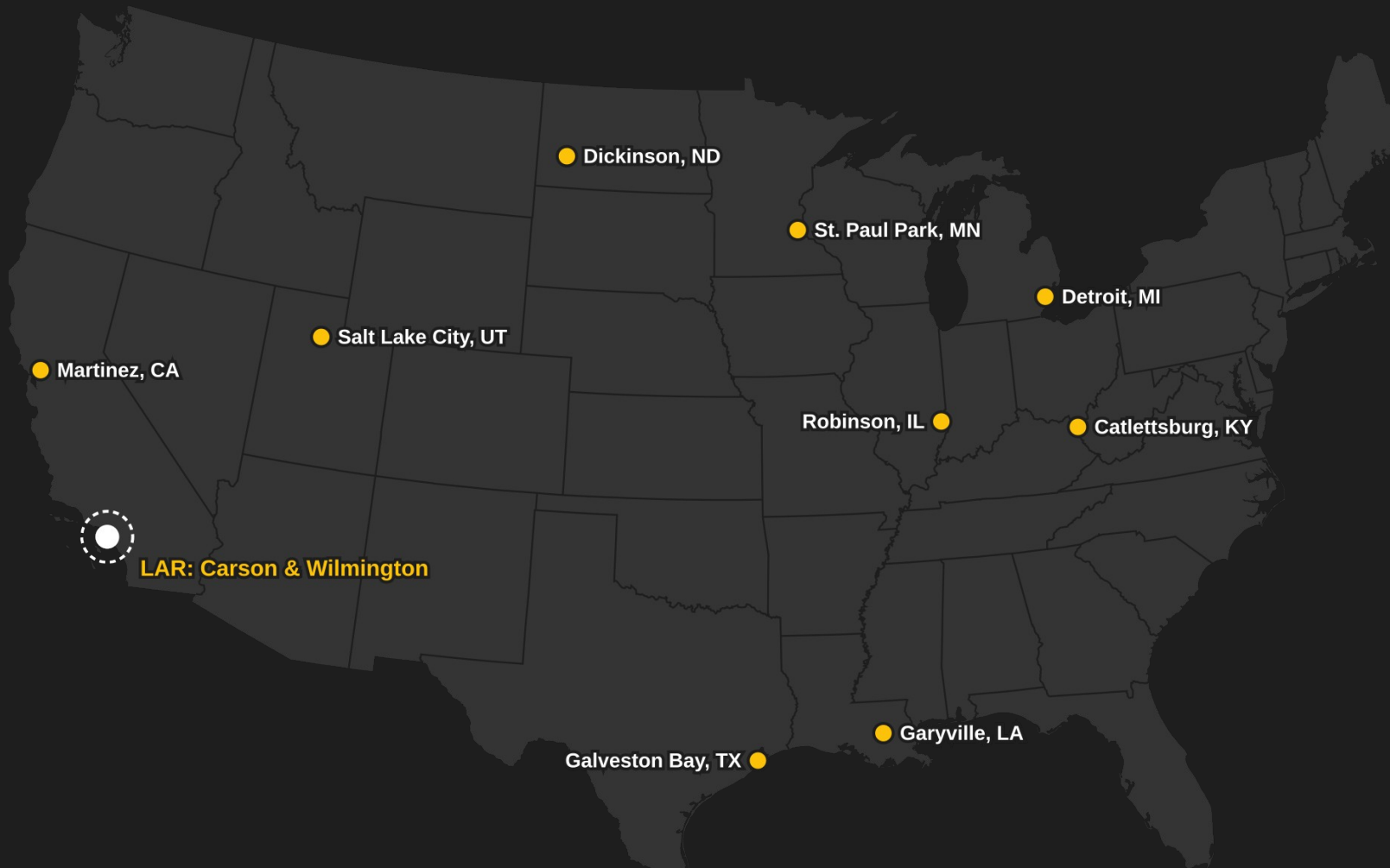


## Data-driven, shared

Durations built from recorded actuals; lessons learned reviewed with Marathon after every job



# Trusted across the MPC system



9

## Marathon refineries with USA DeBusk references

- Named contacts at 8 sites (Galveston Bay on request), listed in the appendix
- Includes Martinez, CA, our West Coast MPC reference
- Crude, coker and vacuum heater experience

# Coker heater case studies & lessons learned

## Marathon Detroit, MI: 70H1 Coker

Decoked September 2024 and September 2026

6

passes, 3 cells

15,156'

coil footage

2

TriMax units

**Scope** All six passes, convection and radiant, pigged simultaneously on two TriMax units.

**Result** Two decokes on the same coker, two years apart: MPC brought us back.

## CHS McPherson, KS: Coker HF-0012

6-pass coker heater, planned decoke, 2025

12,036'

coil footage

6

passes

85 h

pigging

**Challenge** Hard coke held pigs in the radiant section.

**Lesson learned** Stage heavy-duty scraper pigs for the radiant from day one, not as a mid-job change.

# Why USA DeBusk for LAR



## Proven experience

200+ years combined; 9 MPC references, including Detroit's coker



## California-ready

CARB-registered Tier 4 fleet, Southern California footprint



## Resilient execution

Triple-pass TriMax: a spare pump on every 2-pass heater



## Verified clean

4-point verification and joint sign-off on every pass



## Safe by the numbers

EMR 0.48, ISNetworld A, zero fatalities in 3 years





# Thank you

Questions & discussion

Marshall Douglas | Director of Pigging Operations

Jason Harman | Commercial Manager | [jharman@usadebusk.com](mailto:jharman@usadebusk.com) | (713) 839-5961

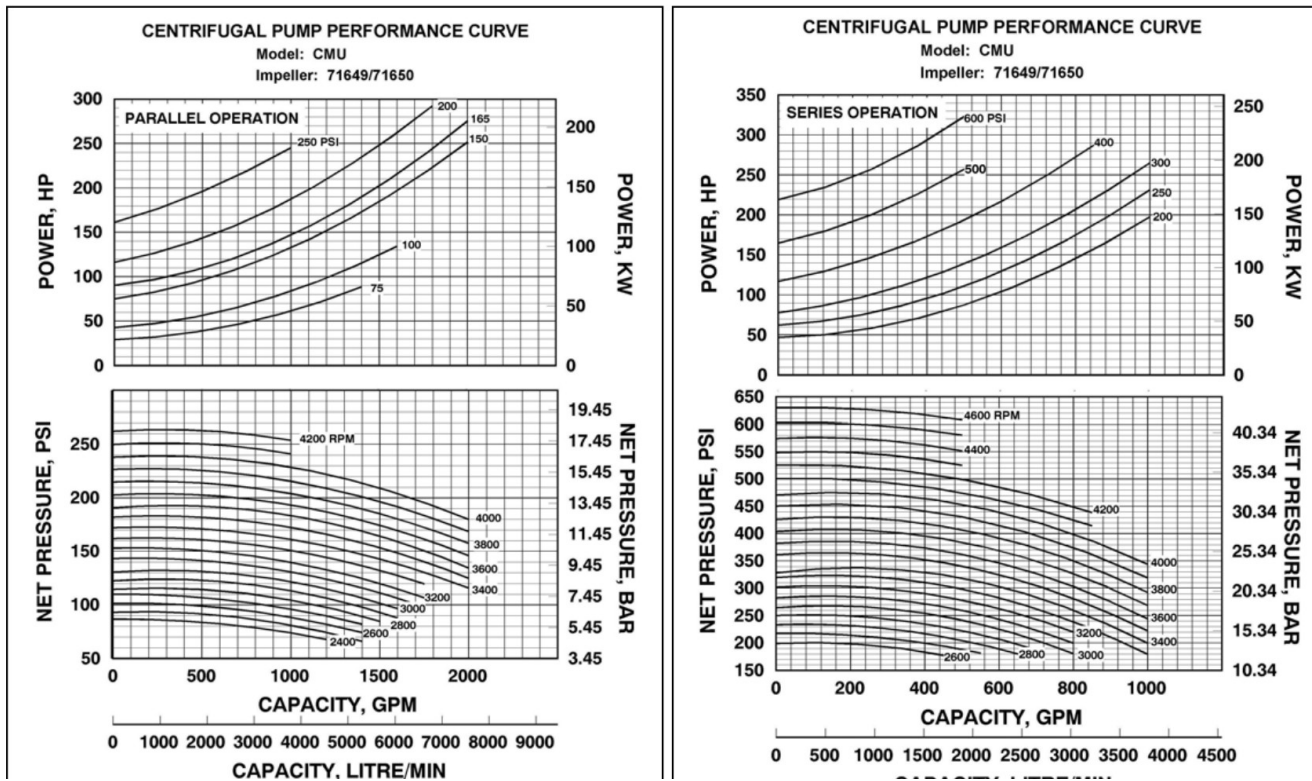
# MPC references

| Refinery           | Contact                | Role                                 | Phone        | Email                              |
|--------------------|------------------------|--------------------------------------|--------------|------------------------------------|
| Catlettsburg, KY   | Eddie Slone            | T/A Planner                          | 606-331-2585 | slone@marathonpetroleum.com        |
|                    | Ryan Rayburn           | TAR / Construction Supervisor        | 606-331-0180 | rdrayburn@marathonpetroleum.com    |
|                    | Ben Holbrook           | Turnaround Planning                  | 606-369-6467 | benlholbrook@marathonpetroleum.com |
| Salt Lake City, UT | Aaron Wiggins          | Maintenance Supervisor / Turnarounds | 801-703-6722 | ATWiggins@marathonpetroleum.com    |
| Martinez, CA       | Kyle Yates             | Project Engineer                     | 925-316-9726 | SYates2@marathonpetroleum.com      |
|                    | Ryan Gorman            |                                      | 952-607-6853 | RMGorman@marathonpetroleum.com     |
| Dickinson, ND      | Tyler Bauer            | TA                                   | 701-516-3456 | tbauer@marathonpetroleum.com       |
| St. Paul Park, MN  | Keith James            | TA Planner                           | 715-441-7986 | TKJames@marathonpetroleum.com      |
| Robinson, IL       | Brian Maus             | TA                                   | 618-544-2121 | bkmaus@marathonpetroleum.com       |
| Galveston Bay, TX  | Available upon request |                                      |              |                                    |
| Garyville, LA      | Shawn Kramer           | TA Planner                           | 225-571-2689 | skramer@marathonpetroleum.com      |
|                    | Chad Childs            | TA Supervisor                        | 281-299-6088 | cechilds@marathonpetroleum.com     |
| Detroit, MI        | Nate Lajiness          | Turnaround Planner                   | 313-297-6166 | nlajiness@marathonpetroleum.com    |

# TriMax pump performance

Waterous CMU two-stage pump, one per pass. Series operation is used for pigging.

Pump Performance Curves



**150–300**

psi normal pigging

**500–550**

psi effective max, after valve losses

**600**

psi rated pump maximum

Manufacturer curves: Waterous Company, form F-2692